

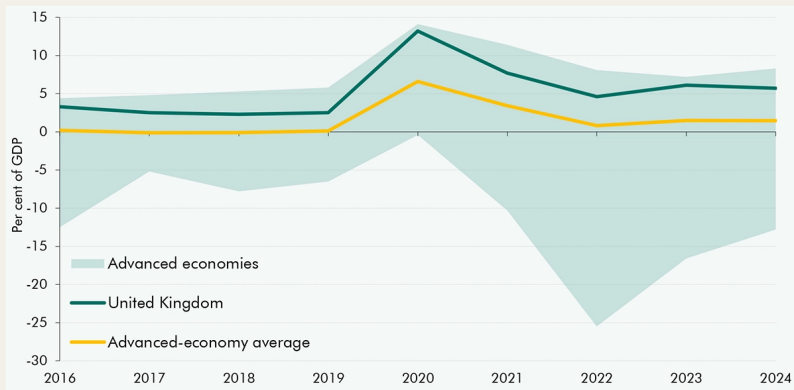


How Much Government Debt is Too Much?

UNIQ - Economics and Management

Sam Blundell, MPhil Economics
University of Oxford (Linacre College)
July 2026

Advanced economies are, on average, borrowing more now than in recent history



GREEK SOVEREIGN DEBT CRISIS, 2010–2012

What we *don't* want to happen

The fiscal deficit of Greece worsened significantly during the 2008–09 financial crisis, when the economy also dipped into a recession. The cost of borrowing for Greece started to rise, and they could not refinance their existing debts.

2010 A **€110bn** emergency bailout to avert default and stop contagion to Spain and Portugal.

2011 A **50% reduction** (haircut) on privately-held bonds, aiming to drag debt down to $\approx 120\%$ of GDP.

Debt sat near 100% of GDP for two decades — then, within months, Greece was locked out of financial markets entirely.

A TALE OF TWO DEBT BURDENS

Why is Argentina forced to slash public spending, whereas Japan plans on increasing it?



Public debt-to-GDP

(2023) 158% → 82%* (predicted 2026)



Public debt-to-GDP

(2020) 258% → 249% (2025)

GROWTH PATHS

The Debt Sustainability Equation

$$\underbrace{\Delta b_t}_{\text{change in the debt-to-GDP ratio}} = d_t + (i_t - \pi_t - g_t) b_{t-1}$$

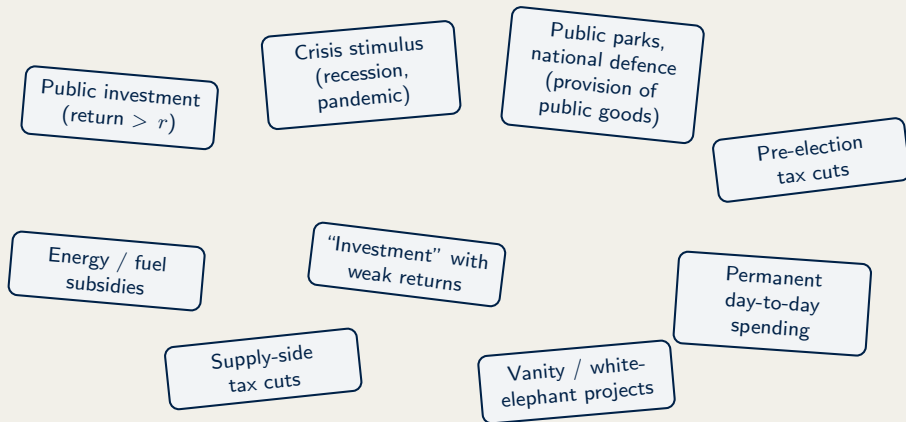
$$\Delta b_t = \underbrace{d_t}_{\text{primary balance}} + (i_t - \pi_t - g_t) b_{t-1}$$

The *primary balance* d_t tracks how much a government spends versus what it raises, *before* interest costs. A primary deficit ($d_t > 0$), where spending exceeds gov. revenue, adds directly to the debt.

Why don't governments just always run a primary surplus ($d_t < 0$)?

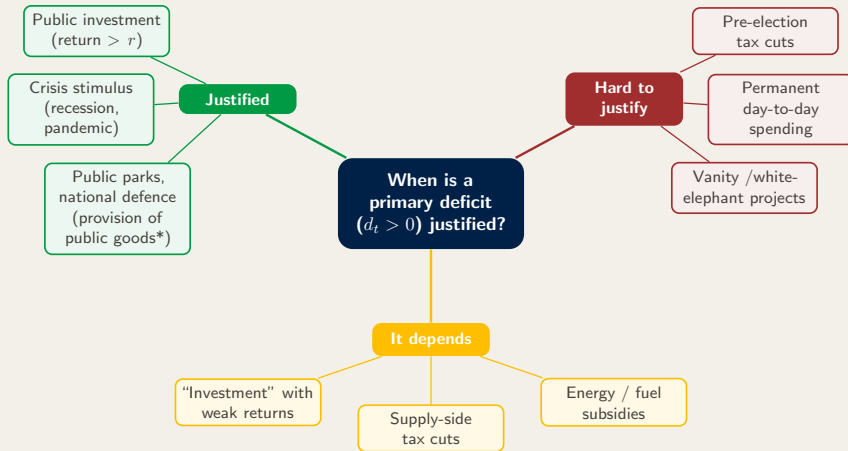
THE PRIMARY BALANCE

Why do sovereigns borrow?



THE PRIMARY BALANCE

Why do sovereigns borrow?



THE PRIMARY BALANCE

Valid Public Expenditure

Efficiency: does it raise future growth g , or smooth a *temporary* shock?

Public goods: does it provide a good that can be:

1. provided to everyone without additional cost, and;
2. cannot (feasibly) be excluded from anyone?

Permanent consumption: if a country is routinely spending more on permanent, predictable consumption (pensions, health care), the gap must be closed to prevent the debt stock rising year-on-year

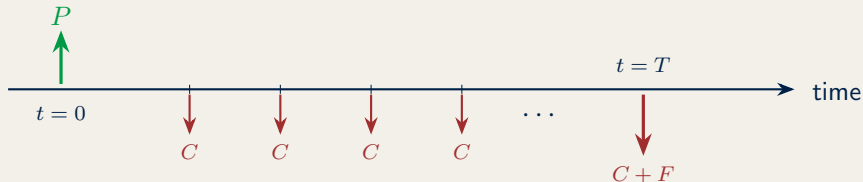
$$\Delta b_t = d_t + \left(\underbrace{i_t}_{\text{interest rate}} - \pi_t - g_t \right) b_{t-1}$$

The nominal *interest rate* i_t sets the price of servicing debt. What determines the rate a government is charged to borrow?

INTEREST RATES

What determines the price of borrowing for a Government?

Governments fund their expenditure via financial markets by issuing **bonds**, a contract that promises a payment (the face value F) at some date in the future (the maturity date), with regular fixed payments in between (coupons C) for some upfront price P



The price of this contract depends on how much investors value these regular payments, adjusted for any risk. **Question:** is this contract risky, or risk-free?

INTEREST RATES

What determines the price of borrowing for a Government?

A bond's **yield** i is the return an investor earns from a bond. It is expressed as a percentage of its current market price, so price P and yield i are inversely related. If more perceived risk $\Rightarrow$ investors pay a lower price $P \Rightarrow$ a higher yield i .

$$i \approx \text{expected policy rate} + \text{term premium} + \text{risk premium}$$

- **Expected policy rate:** the short-term rates which investors expect the central bank to set over the bond's life (this reflects expected inflation and growth).
- **Term premium:** extra compensation for investors holding bonds *long*: money is tied up while rates and inflation may move.
- **Risk premium:** extra return for the chance of *not being repaid* (default).

Further reading: CORE Econ: rates of return, risk & bond prices

$$\Delta b_t = d_t + \underbrace{(i_t - \pi_t - g_t)}_{r_t - g_t \text{ (snowball)}} b_{t-1}$$

Write the real rate $r_t = i_t - \pi_t$, and the bracket becomes $r_t - g_t$. When the real interest rate *exceeds* the growth rate g_t , debt grows under its own weight – even with a balanced primary budget.

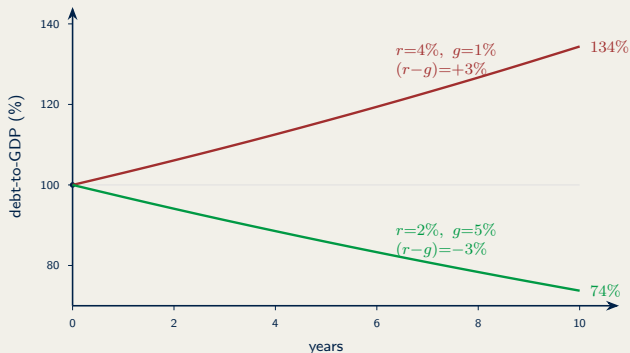
$$r > g$$

Why is debt-to-GDP important?

$$\Delta b_t = \underbrace{d_t}_{\text{primary balance the level}} + \underbrace{(r_t - g_t) b_{t-1}}_{\text{snowball the growth}}$$

- **Level (d_t):** a proportion of GDP added every year, **independent of the stock.**
- **Growth ($(r_t - g_t)b_{t-1}$):** scales with the stock, so it *compounds*.

Set $d_t = 0 \rightarrow$ the path is a race between r and g .



$$r > g$$

Why is debt-to-GDP important?

Economy	Debt %GDP	r %	g %	$r - g$ pp = % of debt	stabilising primary balance % of GDP
India	83	1.5	6.5	-5.0	-4.2
China	90	2.0	4.8	-2.8	-2.5
Japan	235	-1.0	1.0	-2.0	-4.7
USA	120	1.0	2.0	-1.0	-1.2
Germany	63	0.5	0.2	+0.3	+0.2
UK	100	2.0	1.3	+0.7	+0.7
Italy	137	2.5	0.5	+2.0	+2.7
Argentina	80	9.0	4.0	+5.0	+4.0

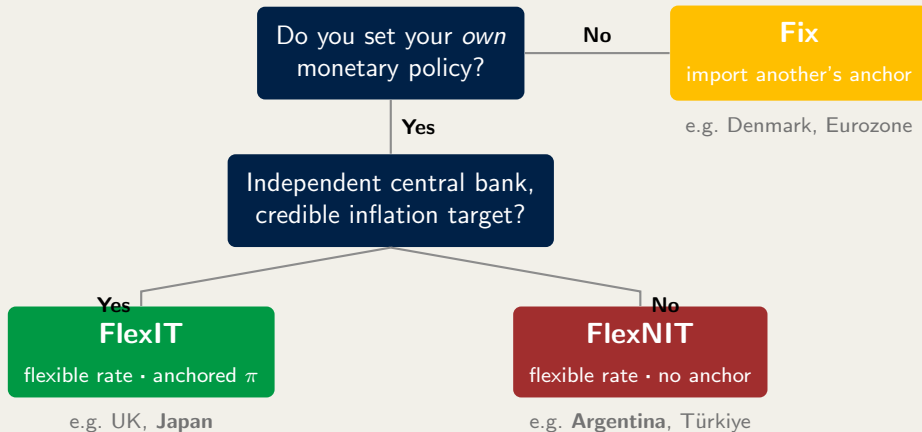
Illustrative ≈ 2025 : g = real GDP growth (IMF WEO); r = real effective rate on government debt (rounded).

$$\Delta b_t = d_t + (i_t - \underbrace{\pi_t}_{\text{inflation}} - g_t) b_{t-1}$$

Inflation π_t erodes the real value of outstanding debt. The monetary regime that sets π_t , and how it interacts with fiscal policy, defines the menu of policy responses available that propagate through every other term.

MONETARY-FISCAL INTERACTIONS

Two questions decide how a country's inflation gets anchored



MONETARY-FISCAL INTERACTIONS

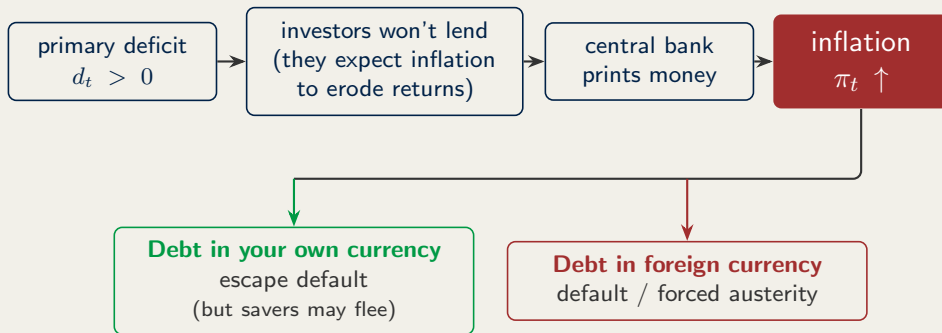
What each regime does

	FlexIT	FlexNIT	Fix
Who anchors inflation?	the central bank	nothing firm	the anchor country
The exchange rate	floats	floats, often sliding	held fixed
After a shock, the exchange rate...	adjusts to cushion debt growth	adjusts to amplify debt growth	cannot adjust

The exchange rate is a shock-*absorber* in FlexIT, a shock-*amplifier* in FlexNIT.

MONETARY-FISCAL INTERACTIONS

Deficits under a non-credible central bank



MONETARY-FISCAL INTERACTIONS

How the regime reaches every term

$$\Delta b_t = d_t + (i_t - \pi_t - g_t) b_{t-1}$$

π_t FlexIT anchors it low • FlexNIT lets it drift • Fix imports it.

i_t credibility shrinks the risk & term premium $\Rightarrow$ lower i_t .

g_t unstable inflation and uncertain pegs drag on growth.

Real rate $r_t = i_t - \pi_t$: a burst of inflation shrinks debt *once* $\rightarrow$ but markets then demand a higher i_t . **No free lunch.**

MONETARY-FISCAL INTERACTIONS

Japan vs Argentina

Japan — FlexIT

own currency (the yen), credible
Bank of Japan
inflation anchored $\rightarrow i_t$ tiny
carries $\approx 250\%$ debt

Argentina — FlexNIT

anchor never held $\rightarrow$ high, volatile
inflation
savers fled to dollars $\rightarrow$ dollarised
can't print dollars $\rightarrow$ forced austerity

$$\Delta b_t = d_t + (i_t - \pi_t - g_t) \underbrace{b_{t-1}}_{\text{existing debt}}$$

Last period's *stock* of debt, b_{t-1} . Its magnitude matters, but so does *who holds it* and *in which currency* it is owed.

Borrow at home or abroad?

Issuing in Own currency (e.g. ¥, £, \$)

Why it stabilises

- **Exchange-rate mitigation:** no currency mismatch → a fall in the exchange rate does *not* raise the real debt burden.
- **No forced default:** in extremes the central bank can always create the currency to pay (cost shows up as inflation, not a missed payment).

The catch: liquidity constraints

- If deep domestic demand from savers (banks, pension funds, insurers) → demand is sticky and slow to flee to other assets → lower yields.
- Low domestic demand from savers → higher yields and crowding out of private borrowers.

Borrow at home or abroad?

Issuing in Foreign currency (usually \$)

Why it tempts

- **Greater liquidity:** access to deep global capital markets and a far larger investor pool than a shallow home market can offer.
- Often a *lower headline rate*.

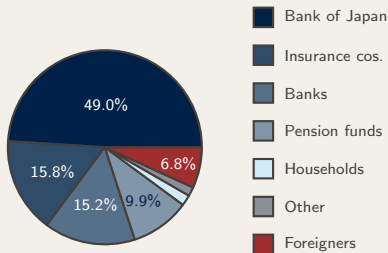
The catch: “the original sin”

- **Exchange-rate risk:** a depreciation of a country's domestic currency against the dollar will inflate the real value of the debt.
- **Balance-of-payments risk:** servicing requires *earning or buying foreign exchange* (exports, reserves, a lender).
- **Rollover / sudden-stop risk:** foreign creditors can flee at once → most likely during a crisis, precisely when it is needed most.

Who owns it? Japan vs Argentina

Japan — JGBs

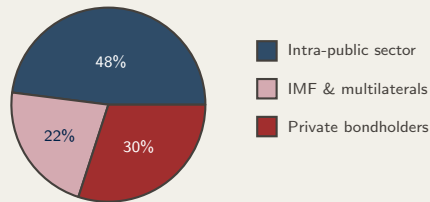
% of JGBs outstanding



≈100% in yen. Half is owed to the central bank;
foreigners hold ≈7%.

Argentina — gross public debt

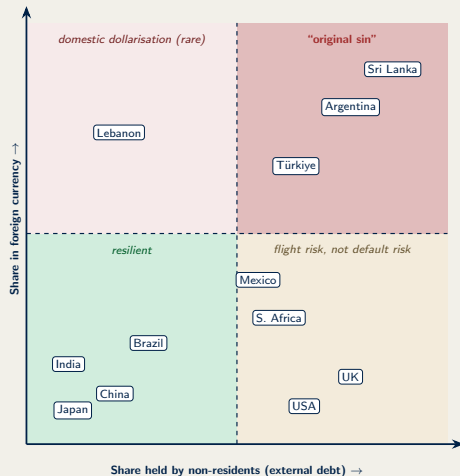
by creditor



≈2/3 in foreign currency. It also borrows heavily “from
itself” — but in dollars it cannot print.

Same trick (borrowing from the public sector), opposite outcome — because of the *currency*, not the ratio.

Who lends, and in whose currency



BRINGING EVERYTHING TOGETHER

The debt sustainability equation in full

$$\Delta b_t = d_t + (i_t - \pi_t - g_t) b_{t-1}$$

Δb_t change in the debt-to-GDP ratio in period t

b_{t-1} debt-to-GDP ratio at the end of the previous period

d_t primary deficit (spending – revenue, before interest)

i_t nominal interest rate on government debt

π_t inflation rate

g_t real GDP growth rate